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المـــدرســـة العليـــا للمـــواصـــلات بتونـــس
ÉCOLE SUPÉRIEURE DES COMMUNICATIONS DE TUNIS

Establishment of a LoRaWAN network using Nemeus equipment

Company:

wevioo



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I would like to thank all the service staff of the welcoming company "Wevioo" who kindly welcomed me as an intern within the firm.

I therefore extend my sincere thanks to my supervisors during the project, Mr. Adel Ghazel who offered me the opportunity to live this professional experience and Mr. Aziz Boulares for all their efforts and invaluable help who offered it to me during the internship.

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General Introduction

The Internet of Things (IoT) refers to the appropriation of the Internet by many varied connected objects. Predictions of the number of sensors show a very strong growth in the years to come for different domains such as home automation, industry or even geolocation, saying it is going to reach more than 50 billion wireless objects in 2022.

Among the protocols created to measure to adapt to the new needs of this IoT, we find the couple LoRa and LoRaWAN:

- LoRa (Long Range Wide Area Network) is the name given to a spread spectrum modulation invented in 2010 and patented in 2012 by the French start-up Cycleo. This Grenoble-based company was bought by the American semiconductor manufacturer Semtech for $ 5 million. It is the proprietary part of the technology.

- LoRaWAN describes the transport protocol that uses LoRa modulation. This layer is open-source and relatively well documented. This recent technology is at the center of the news and of all the attentions since Semtech has found investments to the tune of 50 million dollars and the technology was well received at CES 2017. The competing Sigfox technology has raised 150 million in funds.

Throughout this project will be using an equipment from the brand Nemeus, offered to me by Mr. Adel Ghazel, consisting of two modules MK002-EU, one Gateway MG003-L-EU and two antennas to be linked with the modules or the gateway for establishing connection.

General presentation of the project

Enterprise:

Since its creation in 1998 the WEVIOO group was known under the name of OXIA, FINOXIA and EBSYS which in the search of international development, in a conference 4/02/2016, announced that they changed their names to form one and only group now situated in five countries: Paris, Tunis, Alger, Dubai and Abidjan.

Talking about Tunis it is stationed in " Technopark el Ghazela, Zone Industrielle Chotrana 2 - 2036 Raoued, Ariana, Tunis ".



Figure 1: The head office of Wevioo in Tunis

It should be noted that WEVIOO is an international consulting and technological services group offering technological solutions led by its business and technology experts in more than 30 countries in Europe, North America, Africa and the Middle East, as the PDG Mr. Mehdi Tekaya insisted that Wevioo aims in the medium, and long term to address "a wider international market, to develop closer proximity to our customers, and to offer solutions adapted to the needs of innovation and digitalization of the market."

With a culture of innovation at the heart of its DNA, WEVIOO invests in R&D to develop and implement the best technological projects, aimed at improving the productivity and profitability of companies by providing them with digital innovation solutions that are perfectly suited to their agility, performance and development challenges.

WEVIOO supports its clients in their digital transformation projects, so it is a Customer-oriented group bringing its expertise and know-how in 3 areas: Consulting, Digital and IOT.

Thanks to their experts in different domains, they provide you with all their business and technological expertise in industry, finance, technology and outsourcing, to ultimately help you succeed in your projects.

WEVIOO's human capital management policy revolves around career management allowing for planned and continuous professional development. Wevioo invests in the development of its employees by offering structured, diversified training courses adapted to the needs and expectations of the market.

▶ A real identity and a real corporate culture.

▶ An environment and pleasant working conditions promoting the well-being of employees.

▶ Teams of professionals, and the best tools dedicated to Human Resources management.

▶ Attractive and scalable compensation management in line with career management.

Project Presentation:

What is IOT?



Figure 2: the different types of connected objects of IOT

The IoT is the acronym for Internet of Things (Internet of Objects in French). The term IoT first appeared in 1999 in a speech by Kevin ASHTON, a British engineer. It was used to designate a system where physical objects are connected to the Internet and also capable of creating and transmitting data in order to create value for its users through various services (aggregation, analytics, etc.).

According to the International Telecommunication Union, the Internet of Things (IoT) is a "global infrastructure for the information society, which provides advanced services by interconnecting objects (physical or virtual) using existing or evolving interoperable information and communication technologies".

The Internet of Things, sometimes written IoT or IOT (Internet of Things), refers to all the infrastructures and technologies put in place to make various objects work through an Internet connection. We are then talking about connected objects. These objects can be controlled remotely, most often using a computer, smartphone or tablet thus presenting a kind of bridge between the physical world and the virtual world.

Thus, the term Internet of Things includes all physical objects and devices that have a digital identity (IP address, smtp, http protocols, etc.). They can be everyday objects ubiquitous in homes (television, refrigerator, washing machine, heating system, electric garage door), more complex devices or systems such as vehicles (planes, autonomous cars). and the lighting of a city.

Thanks to information and communication technologies, a bridge between virtual and physical objects is made thus the ability to change the behavior of an object to match that of other objects.

An IoT solution is built around 5 essential components which are:

- Objects (sensors).
- The network (connectivity).
- Data.
- Information.

- Operating applications.

What is LORAWAN?

The LoRa network, also called LoRaWAN, is a data network dedicated to IOT sensors. It provides small-scale data transmission over large distances - up to 15 km, while consuming very limited power, which makes it possible to use sensors on small batteries.

Developed by Semtech. It is based on spread spectrum modulation techniques derived from chirp spread spectrum (CSS) technology.

The LoRa Alliance is the fastest growing technology alliance.
It is a non-profit association with large number of members (more than 500 member companies), working to enable large scale deployment of Low Power Wide Area Networks (LPWAN) IoT through the development and promotion of the LoRaWAN open standard.

Architecture

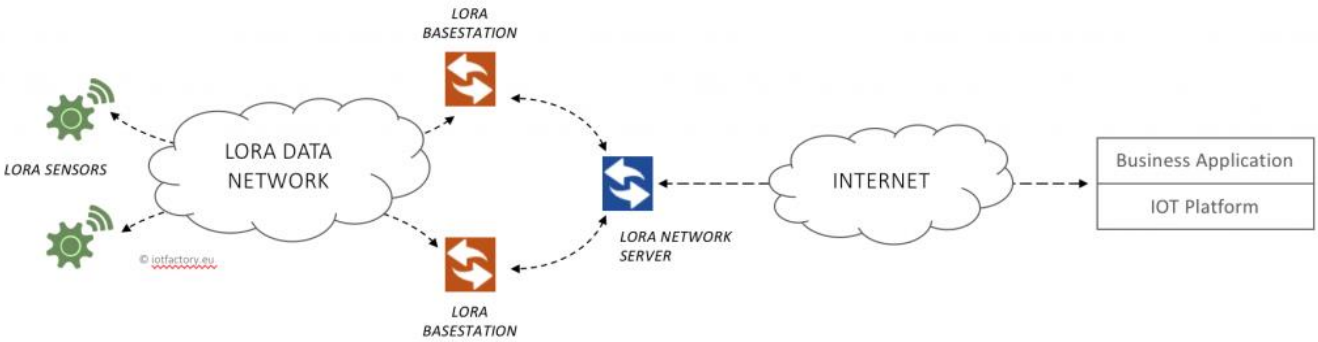


Figure 3: LoRaWAN architecture

Gateways (LoRa Base station) relay messages between end-devices and a central network server.
The gateways are connected to the network server via standard IP connections and act as a transparent bridge, simply converting RF packets to IP packets and vice versa. The wireless communication takes advantage of the Long Range characteristics of the LoRa physical layer, allowing a single-hop link between the end-device and one or many gateways. All modes are capable of bi-directional communication.

Classes

- *Class A - Lowest power, bi-directional end-devices:*

The default class which must be supported by all LoRaWAN end-devices, class A communication is always initiated by the end-device and is fully asynchronous. Each uplink transmission can be sent at any time and is followed by two short downlink windows.
Since the end-device is able to enter low-power sleep mode for as long as defined by its own application: there is no network requirement for periodic wake-ups proving class A to be the lowest power operating mode, while still allowing uplink communication at any time.

- *Class B – Bi-directional end-devices with deterministic downlink latency:*

Class B devices are synchronized to the network using periodic beacons, and open downlink 'ping slots' at scheduled times. This provides the network the ability to send downlink communications with a deterministic latency, but at the expense of some additional power consumption in the end-device for battery powered applications.

- *Class C – Lowest latency, bi-directional end-devices:*

In addition to the class A structure of uplink followed by two downlink windows, class C further reduces latency on the downlink by keeping the receiver of the end-device open at all times even when the device is not transmitting (half duplex), making the network server able to initiate a downlink transmission at any time on the assumption that the end-device receiver is open, so no latency.
The problem is the power drain of the receiver (up to ~50mW) and so class C is suitable for applications where continuous power is available.

=>For battery powered devices, temporary mode switching between classes A & C is possible, and is useful for intermittent tasks such as firmware over-the-air updates.

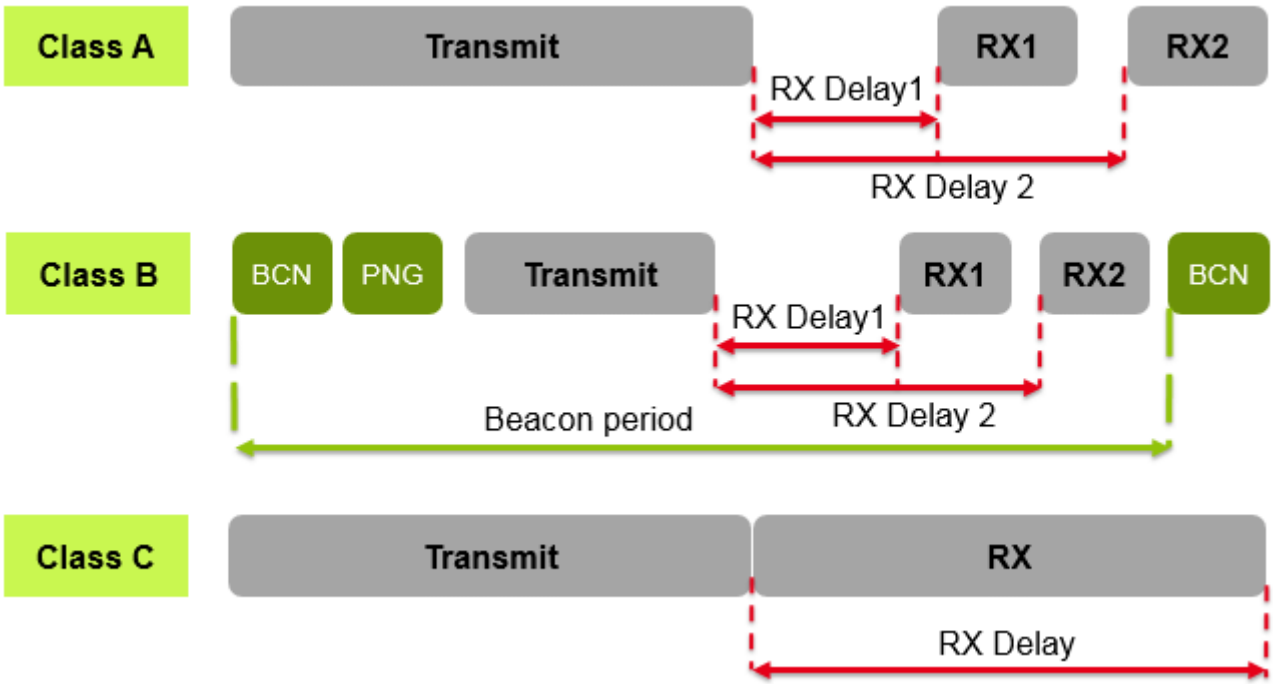


Figure 4: Detailing the way of connection of the three LORAWAN classes (A, B and C)

Conclusion:

In this first chapter, we presented a general description of the host organization, Wevioo, and subsequently we presented the framework of our project in which we will be establishing the connection between Nemeus modules and Gateway and cloud to create a LoRaWAN network.

Used equipment and technologies

Introduction:

After establishing the context of the project, I am going to talk about the used equipment and technologies during the project of my internship.

Nemeus:

Modules:



Figure 5: Module MK002

The MK002-xx-EU is the evaluation kit of the MM002-xx-EU module. Plugged onto a PC solution, this USB key allows to manage bi-directional communications through:

- LoRa™/FsK RF layers between USB key and any solution embedding MM002-xx-EU module
- LoRa™ WAN 1.0.2 EU Class A & C bidirectional protocol with Public or Private networks
- SIGFOX™ bidirectional protocol

Important Features:

- Modulation: LoRa™ / SIGFOX™/ FSK
- Controlled by AT commands through USB interface
- Good buildings penetration
- Long Range protocol stack software (SIGFOX™ / LoRa WAN 1.0 EU Class A & C)
- RoHS conform
- Pre-Certified EN 300-220 / RED directive

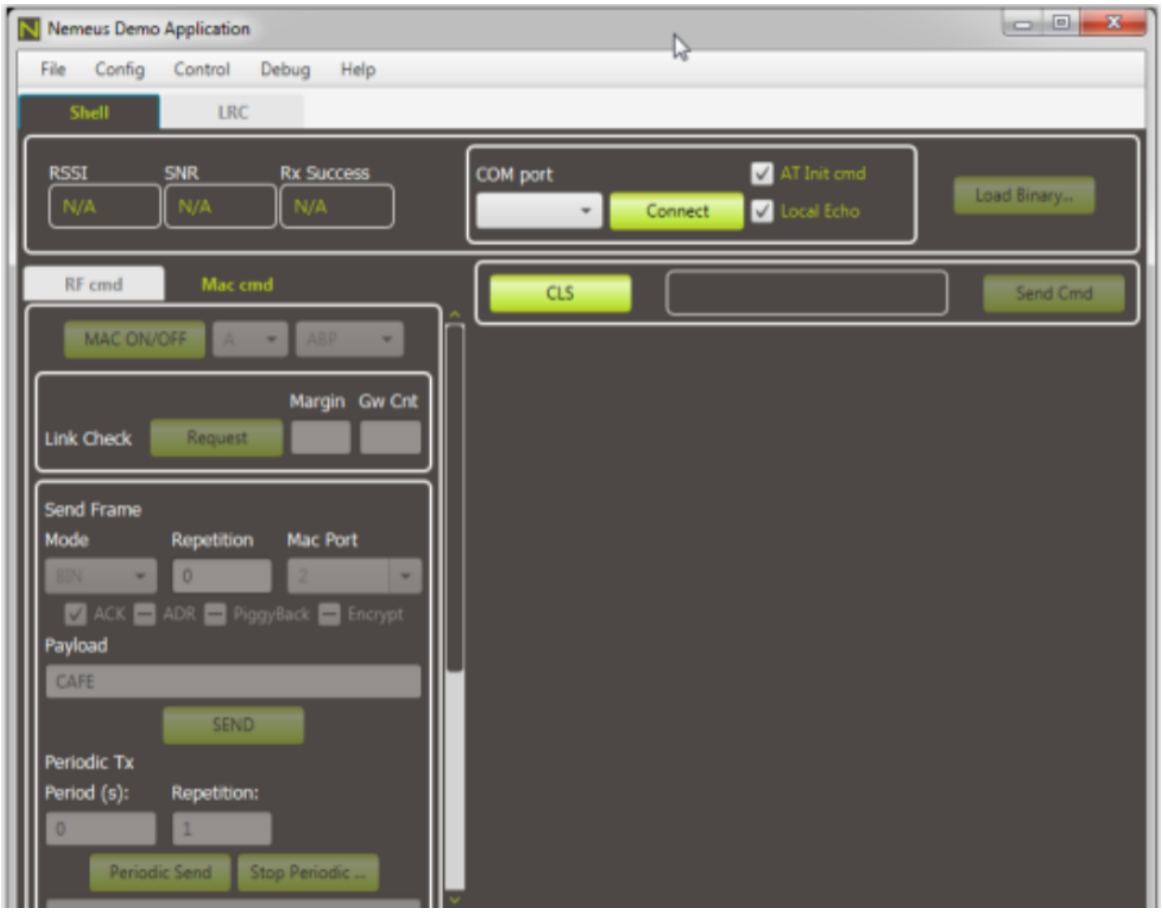


Figure 6: Module JAVA Application

Requirements:

- JAVA Runtime Evaluation 8 kit is installed onto the computer.
- The file “o_NemeusATKv3.01.jar”

Execution:

- First on a Command DOS windows go to folder where the file “o_NemeusATKv3.01.jar” resides, then execute the command “java -jar o_NemeusATKv3.01.jar”.

During execution, the JAVA application records information in a log file which is created in same folder of the .jar file at the time this one is started. The generated log file stays the same during all the JAVA application execution.

Gateway:

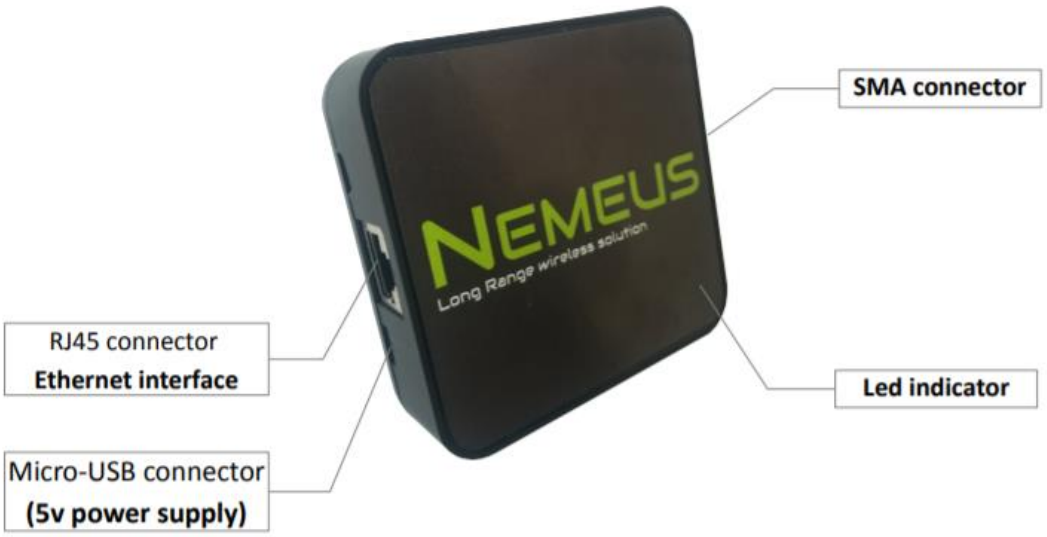


Figure 7: Gateway MG003

The MG003-L-EU is a pico-gateway dedicated to long-range indoor communication for low power wireless devices operating on ISM 868Mhz unlicensed band (LoRa modulation). With an embedded LoRa WAN Class A & C 1.0.2 EU server able to manage up to 400 devices (no limit in non-server mode) and good radio performances, it is a good solution to deploy easily an all-in-one local LoRa-WAN private network.

Important Features:

- Dimension: 75 x 75 x 25mm
- External antenna provided (Gain=2dBi)
- Supply with 5V/0,5A from micro-USB cable
- Ethernet network connection to customer applicative server
- 2 modes: Packet forwarder and LoRa Wan Class A & C 1.0.2 EU embedded server
- 8 LoRa channels solution
- Management of one hundred LoRa devices
- Certified LoRaWAN 1.0.2 class A & C network
- Indoor version
- Remote configuration through remote JAVA application



Figure 8: Gateway JAVA application

Requirements:

- JAVA Runtime Evaluation 8 kit is installed onto the computer.
- The file “o_NemeusGatewayv1.54.jar”

Execution:

- First on a Command DOS windows go to folder where the file “o_NemeusGatewayv1.54.jar” resides, then execute the command “java -jar o_NemeusGatewayv1.54.jar”.
- Nemeus Java Gateway Application Used to manage the gateway via Network interface
- During execution, the JAVA application records information in a log file which is created in same folder of the .jar file at the time this one is started.

Configuration:

- DHCP folder: DHCP configuration and parameters
- DNS folder: DNS IP addresses
- NTP folder: Simple Network Time Protocol configuration
- SNET folder: Data & Security Server connection parameters

- FWD folder: Packet Forwarder parameters
- RADIO folder: LoRa/FsK Radio(s) configuration
- RX FILTER folder: Rx Frame Filter
- UPDATE folder: Gateway information

Antenna:



Figure 9: Omnidirectional Antenna

Radiates Wi-Fi / Bluetooth / wireless signal high and low enough to reach 2 levels of a building.

- As client antenna: Connect to access points in any direction. Portable size: Easily fits in a laptop case.
- As access point antenna: Provides signal to clients in all directions (360-degree horizontal beamwidth)
- Mounts directly onto RP-SMA (female) connector of access point, wireless bridge, client, or booster
- Bottom section of antenna has hinged joint that bends up to 90-degrees. Can be used either straight or with joint-bend.
- VSWR < 2.0
- Gain 2dBi ± 0.7dBi
- Efficiency > 81% @ 2400 ~ 2500MHz
- Impedance: 50 Ohms
- Compatibility with wireless standards & application

TTN (The Things Network):



Figure 10: Logo LoRa Alliance



Figure 11: Logo TTN

The Things Network is a member of the contributor of the LoRaAlliance as it works to enable low power Devices to use long range Gateways to connect to an open-source, decentralized Network to exchange data with Applications.



Figure 12: Simplified network connection

Thankfully the offered solution is easy to deploy and maintain with the help of the provided documentations and the active community to support different existing problems.
We provide a set of open tools and a global, open network to build your next IoT application at low cost, featuring maximum security and ready to scale.
It is mainly used for tests to try and improve an existing LoRaWAN or build a new one since it has many limitations.
The Things Network professional offer is named The Things Industry to help industries build a secured, scalable and a robust network using long range, low power and a cost effective equipment.



Figure 13: Logo TTI

Chirpstack:



Figure 14: Logo ChirpStack

The ChirpStack open-source LoRaWAN Network Server stack provides open-source components for LoRaWAN networks. Together they form a ready-to-use solution including a user-friendly web-interface for device management and APIs for integration. All components are licensed under the MIT license and can be used for commercial purposes.

The only problem is the complexity of maintaining and developing the solution using the ChirpStack cloud: It does provide documentation and a wide community but for professional and free use It is complex.

Feature highlights:

- Class-A support
- Class-B support
- Class-C support
- Adaptive data-rate
- Live frame-logging
- Channel (re)configuration
- Multi-tenant
- APIs and integration
- LoRaWAN 1.0 and 1.1 compatible
- And much more……

Conclusion

In this part I spoke about the used equipment from Nemeus which were provided by Mr. Adel Ghazel then I continued with the used technologies offering the LoRaWAN solution.

Performed Tasks

Introduction:

My project was divided in a steps to follow to ultimately finish it, this part is going to enumerate those steps and define each part.

Research:

As it is clear from the name this is about the research done to find the possible open source providing the LoRaWAN solution.
During this phase I did find a lot of various existing servers in the market from which we will name:







All this existing servers offers the LoRaWAN solution and is open source yet not all are helpful to my project since some are not free.
Thus eventually we chose Chirpstack as our LoRa sever even if it may be complicated but it was the only one found to be free.

Establishing connection between modules:

First step of configuring the modules of Nemeus was establishing a connection between the modules after connecting both of the modules to different Computers and have these modules equipped each with one antenna then activating the JAVA application of those modules.

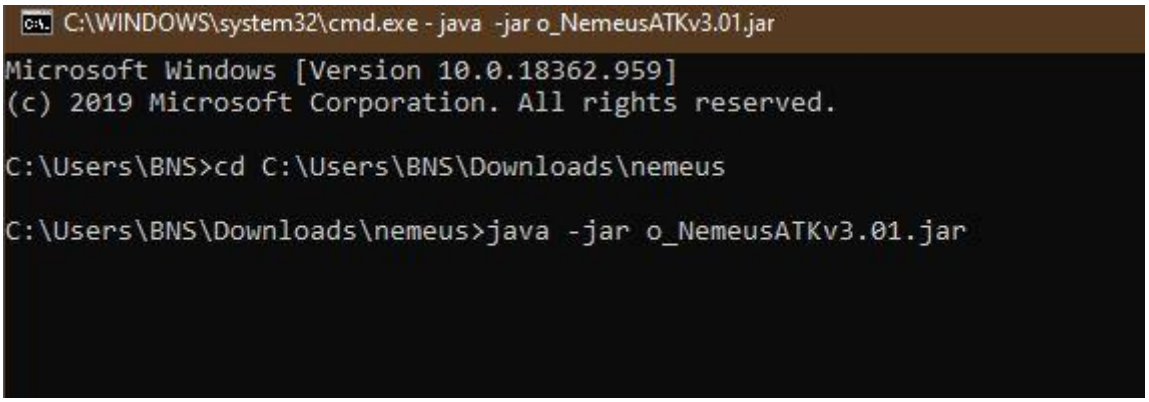


Figure 15: Used command to run the JAVA app

After configuring the parameters of the application a connection is established between them which is defined by the numerous command displayed on the application's console.



Figure 16: Port of the connected module

In the console the user can send commands after writing them in the upper part then pushing the “Send Cmd” button



Figure 17: The console of the JAVA application

In this part to get both of the module to connect to each other one should be put as receiver as the other should be a transmitter provided that the RF option is set on.

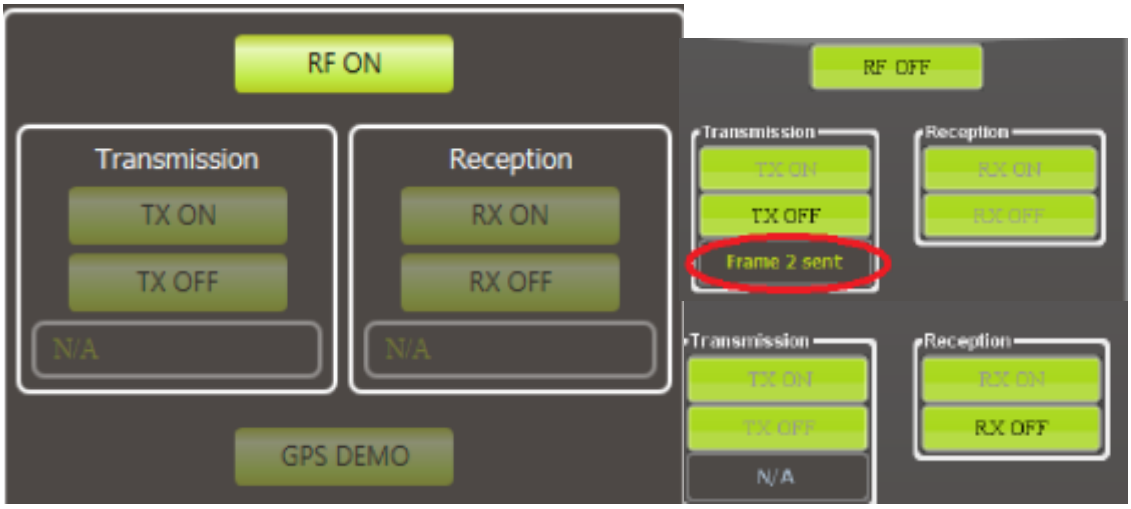


Figure 18: RF configuration of modules

As such we can see the sent traffic on the console.

Establishing Connection between modules and Gateway:

For this part since I do have only to antenna the connection was only between one module and a Gateway and for the connections I am using only one pc.
So first the Gateway must be connected to a source of power and another source of internet connection using an Ethernet cable, after waiting for a little time the gateway will have an IP address which is going to be used to establish connection between the JAVA application and the Gateway.

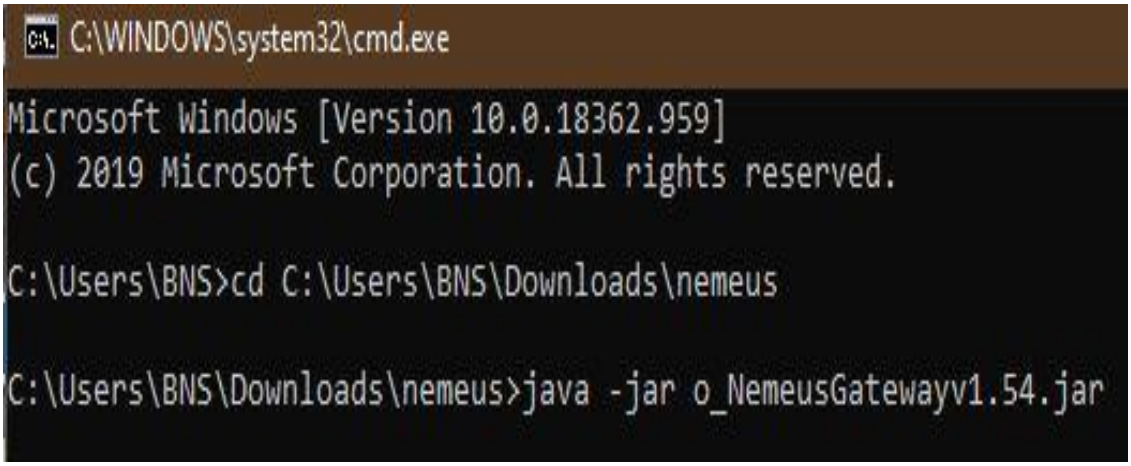


Figure 19: Used Command to activate Gateway JAVA application

After activating the application logs will be displayed on the command line until closing the application.

To find the IP address of the Gateway I had to scan the network for all used IP address with the “Fang” application and after finding it I put it in the JAVA application to establish connection between them like in the figure using the IP: 192.168.0.11



Figure 20: Connection established



Figure 21: Fing Android app

Now after activating on of the modules on the Computer like in the previous section yet this time I am going to use the LoRA™ WAN option because the connection between gateway and module is not through RF configuration.

With this configuration the user can configure the use of Class A or C in LoRAWAN connection and send a specified number of payloads (used

"CAFE" by default) to test the connection between whatever used equipment and the module.

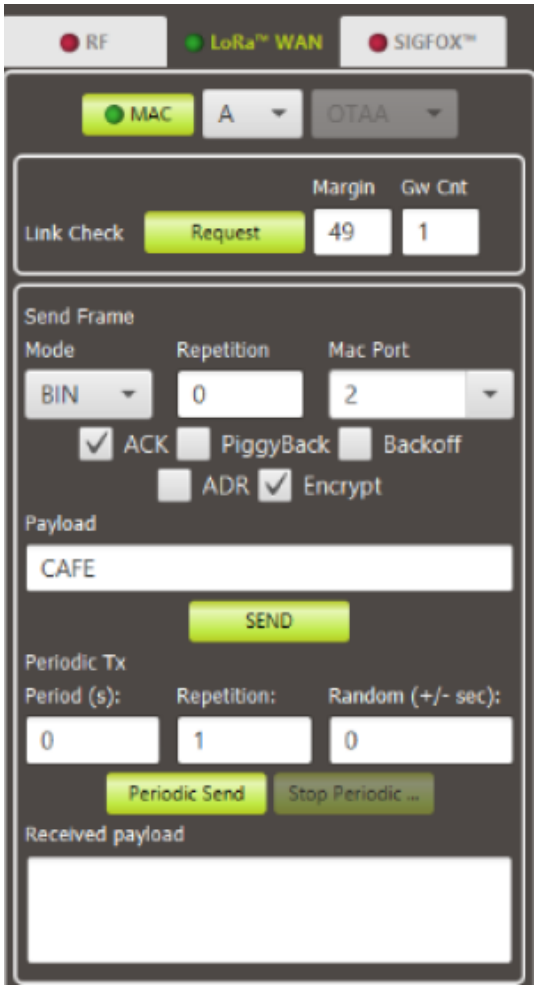


Figure 22: The LoRaWAN option

Now it is time to connect the module to the gateway, to do that I have been provided with a txt document registrating all information on a many module from them we will select the used one by idetifying its ID.

{"id":5469,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5470,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5471,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5472,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5473,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5474,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5475,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
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{"id":5477,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5478,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk
{"id":5479,"sw":"nemeus-mm002-master16W18_08-modem","hw":"mm002 v2.1","customer":"esigelec-170824","comments":"mk

Figure 23: Information about the modules

We selected the ones with the ID=“5473” and the ID= “5477”.
Now we go to adding the configuration of the modules to the Gateway app:
First push the “Device List” and then “Add device” to configure the modules identity.

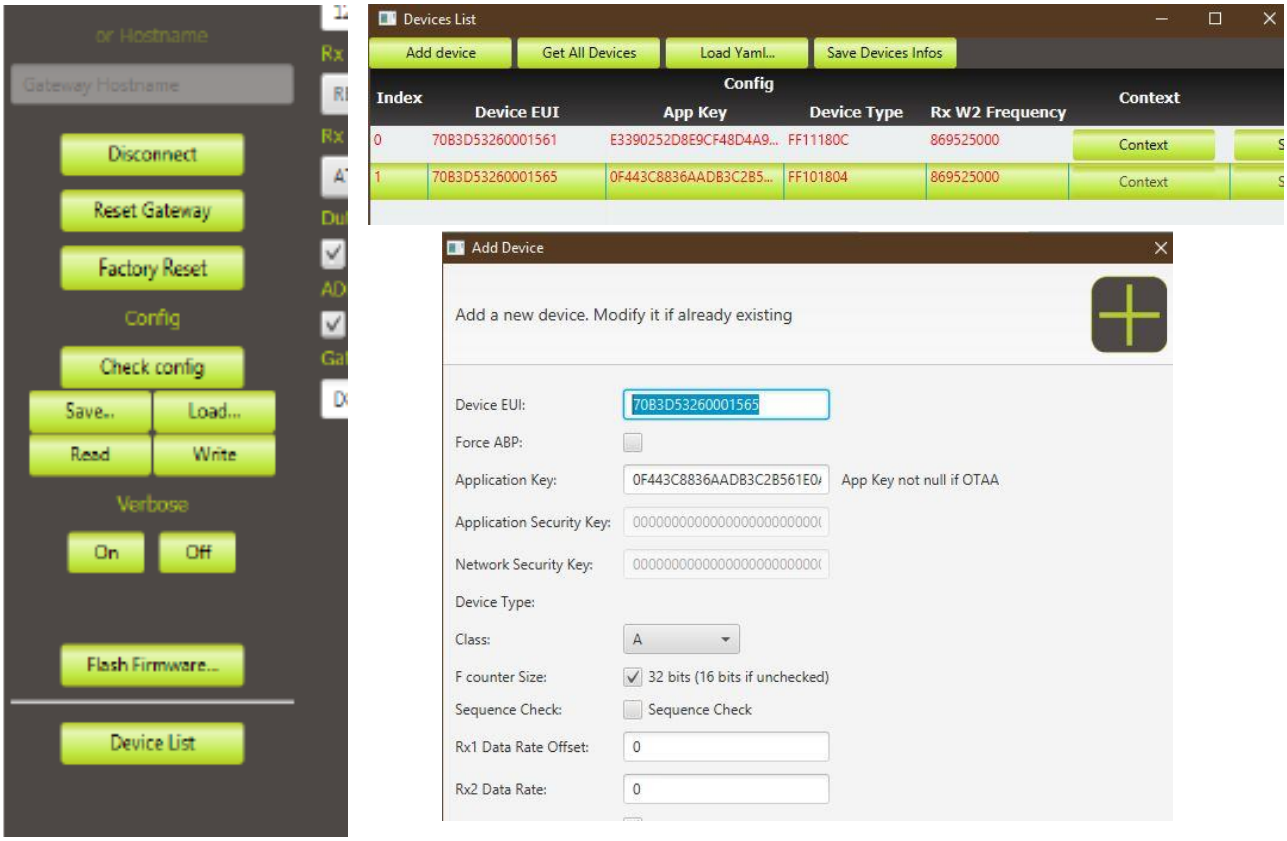


Figure 24: The configuration of the module

Or the option “Get All Devices” which can help detect the existing modules in the network.
Then we go to the Nemeus module JAVA application to establish the configuration of the used Gateway.

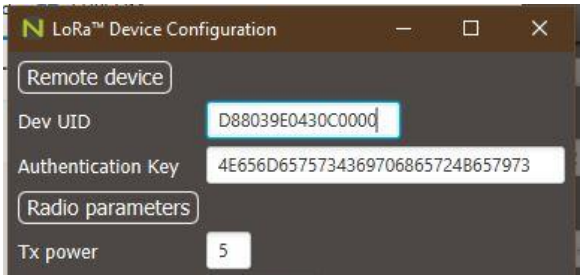


Figure 25: The ID and Authentication key of the Gateway

Thus now we have established the connection between one of the modules and the Gateway.

Establishing the same connection through the TTN

It is true saying it is the same connection may be wrong but here we pointing to the name of the equipment interconnection yet this time is through the TTN server
First we create an account on the TTN site afterwards we go to the option Console

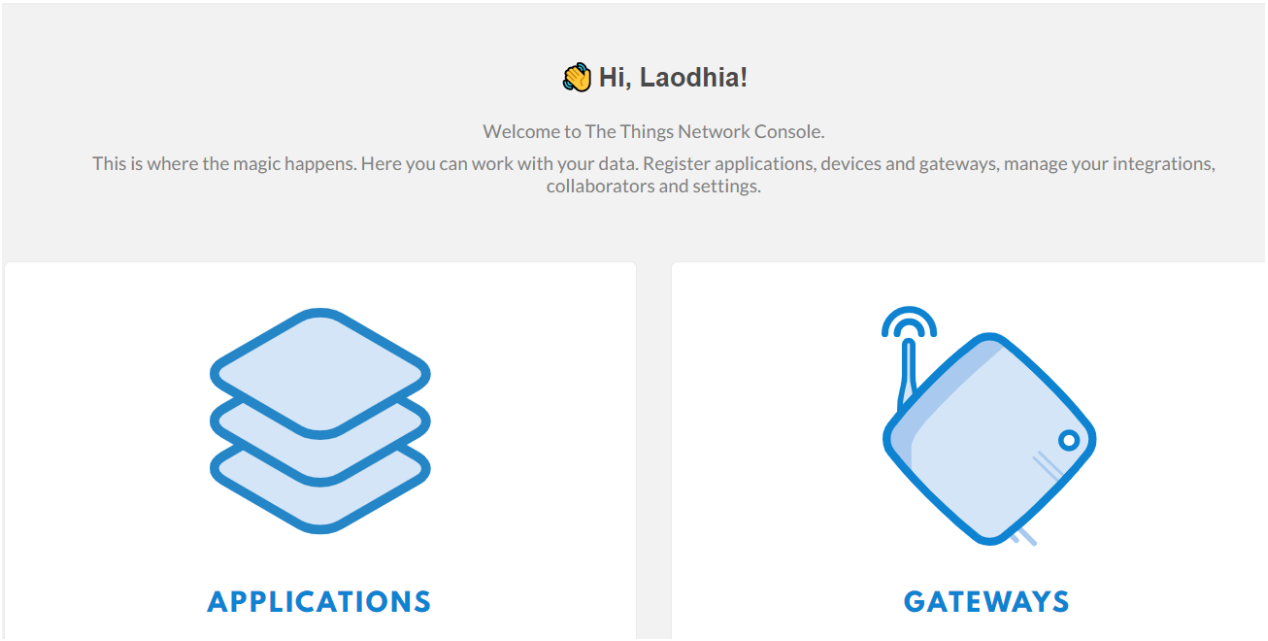


Figure 26: The console option

Now we choose whatever we want in my case I chose Gateway that is where you push on your Gateway configuration

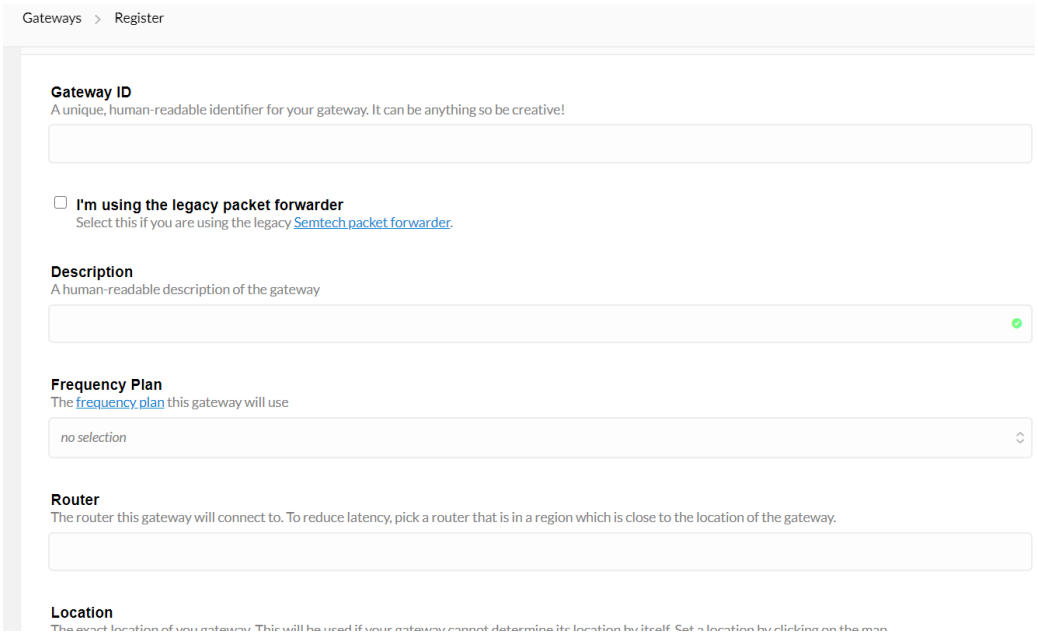


Figure 27: Gateway configuration

Afterwards the gateway “Demo” will appear as shown in the next figure:

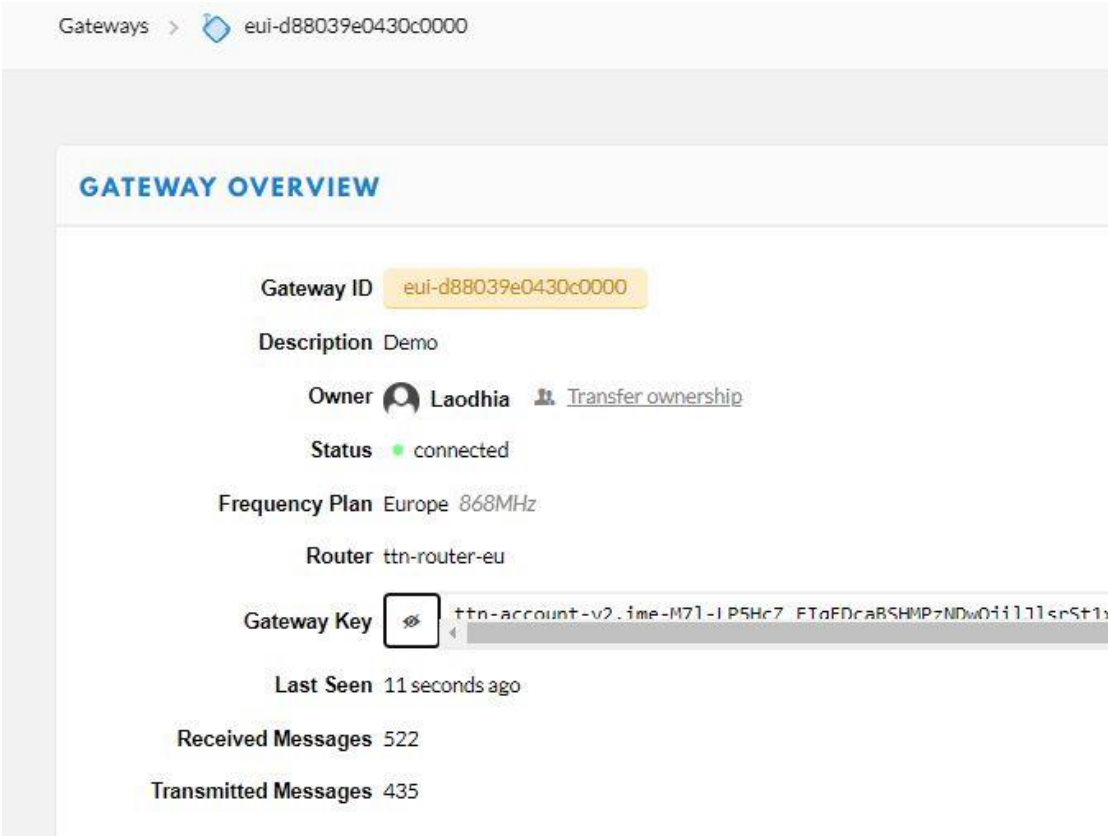


Figure 28: Gateway Overview

Then we go to the application part and choose “Add application”:

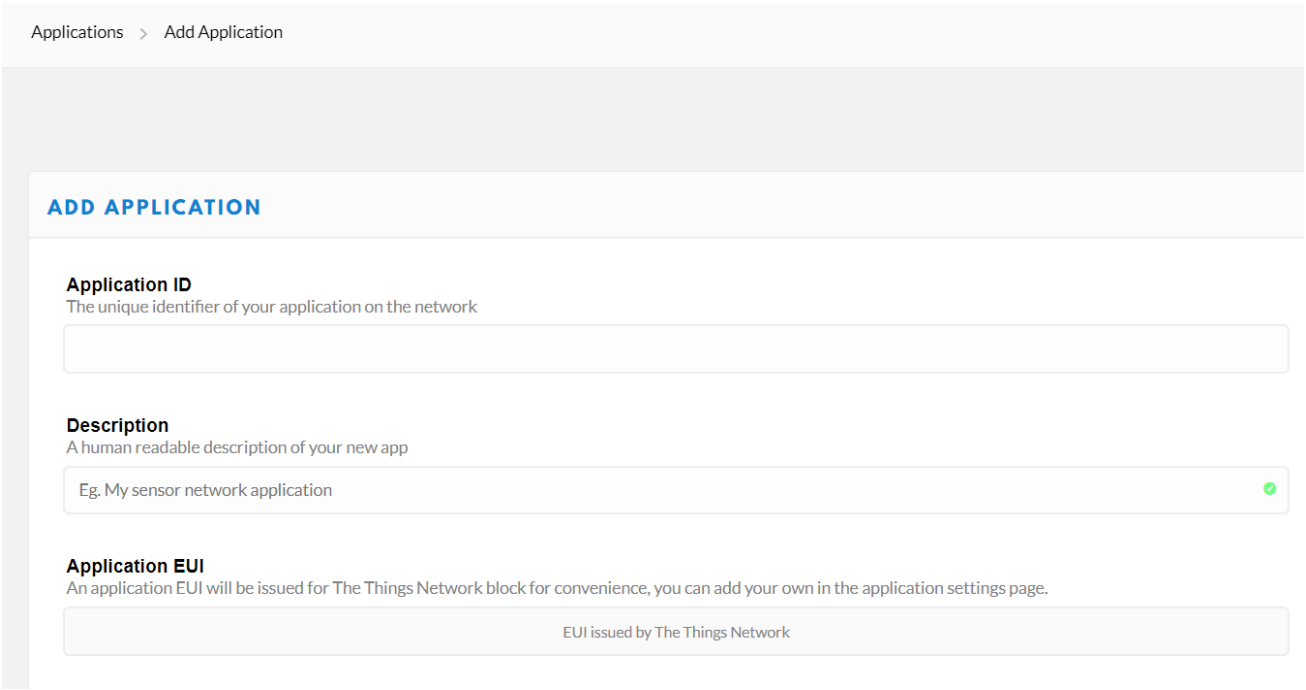


Figure 29: Application configuration



Figure 30: Application “demo” status

Most configuration can be using the default values then the configuration of the modules needs to be added .

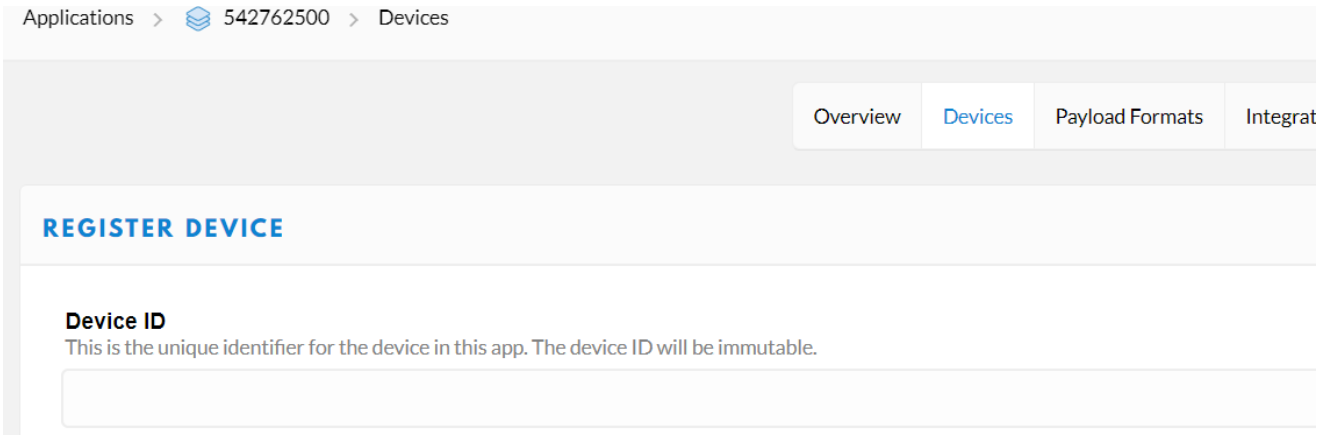


Figure 31: Configuration to register device

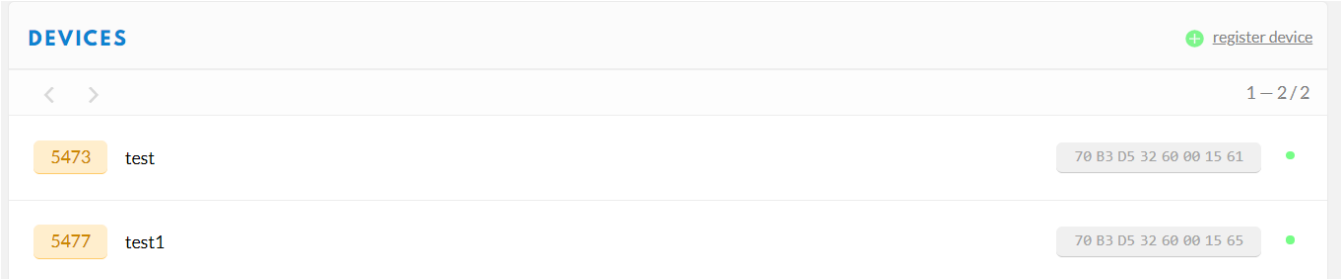


Figure 32: The status of the registered devices

Now is the part where the application ID needs to be added to the module



Figure 33: Connection through OTAA to TTN Application

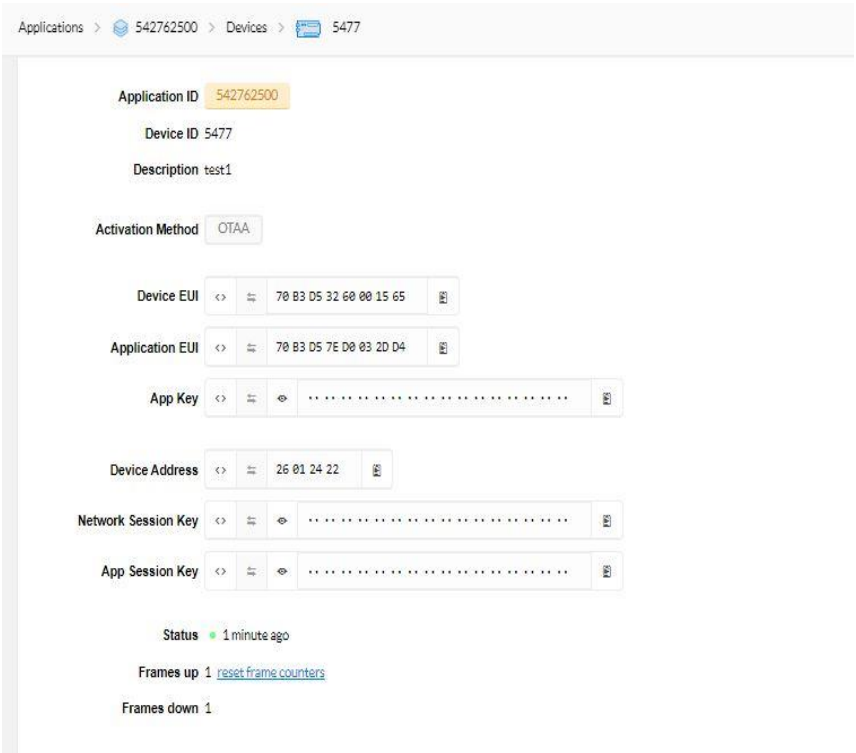


Figure 34: Device “5477” status

Finally we can say that we did establish the connection through the TTN application on the cloud now we can test the sending and the receiving of the packets.

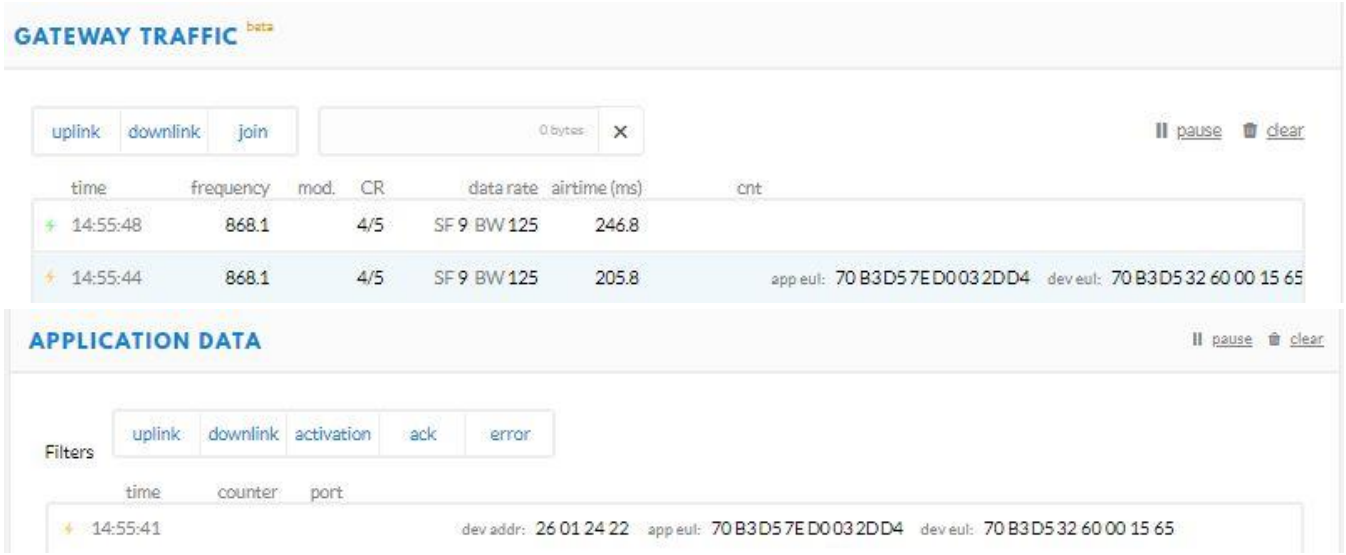


Figure 35: Gateway and Application traffic

Both figures show the effect of a join between the gateway and the device with the shown packets.

In the next figures we see the sent packets traveling through the equipments:

Note: The next figures were taken after the packet limit is reached thus explaining the problem of no payload.

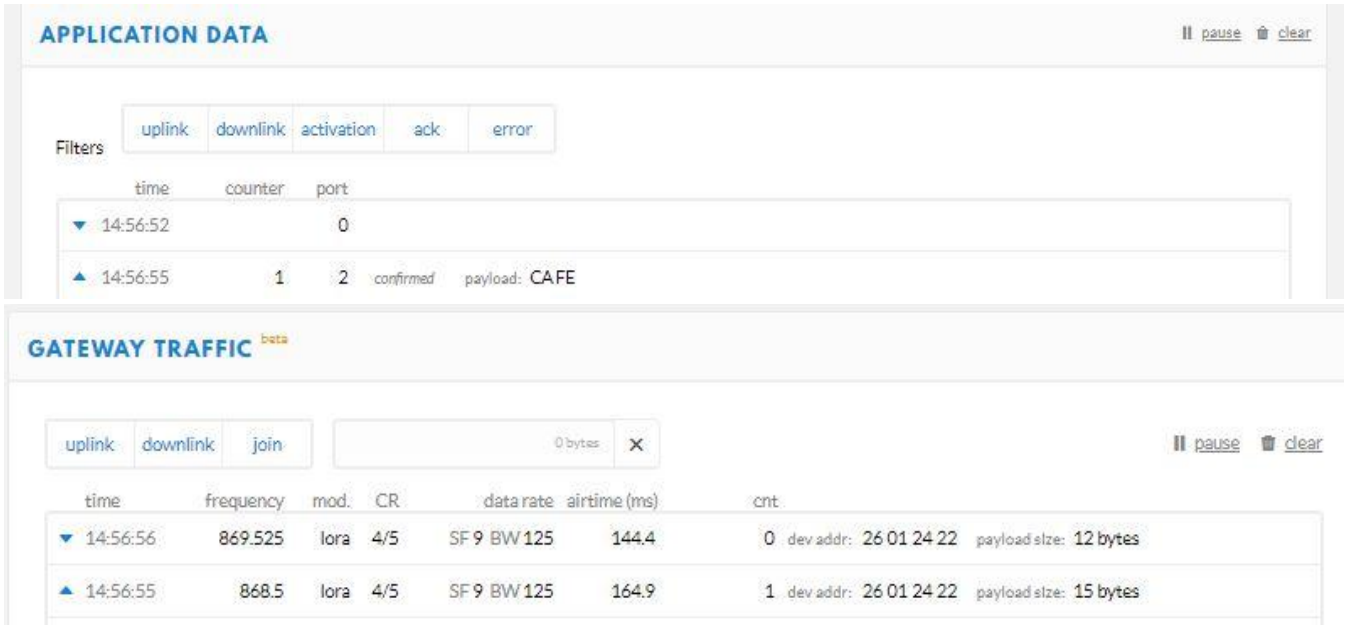


Figure 36: The sent packets (Payload : “CAFE”)

Establishing the same connection through the Chirpstack server:

In this part I had a long time to configure and install Chirpstack on my PC with the problem being its complexity so first as I have a windows OS I chose the configuration using “Docker”, “PostgreSQL”, “Redis” and “Mosquitto”.



Figure 37: The logos of used application

After passing a long time understanding their way of functioning and establishing the connection in between I understood the way it works which is but installing and configuring each on windows there is a default code on “Github” to establish their interconnection yet I had some mishaps thus I turned to the use of Ubuntu.
With the advice of my supervisor I chose to configure a dual boot with Ubuntu on my machine afterwards the install and configuration is easier since it is a series of Linux commands making it more understandable and the configuration on the specified files, this was with the help of some tutorials and the documentation provided even if it is sometimes not clear. Files on “etc.” folder holding the configuration of ChirpStack, with those we insure the activation of all the ChirpStack files with the configuration related to the provided Nemeus equipment

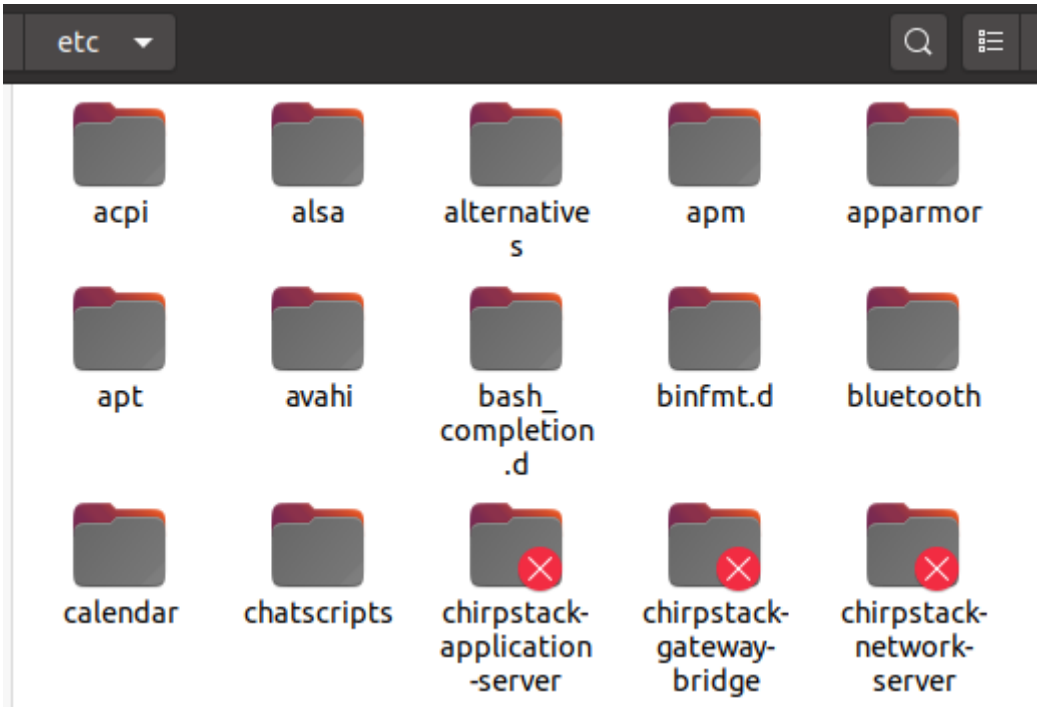


Figure 38: the installed files

Now we will be activating all installed configure:

```
root@kol-GL65-Leopard-10SDR:/home/kol# sudo systemctl start chirpstack-gateway-b
ridge
root@kol-GL65-Leopard-10SDR:/home/kol# sudo systemctl status chirpstack-gateway-
bridge
● chirpstack-gateway-bridge.service - ChirpStack Gateway Bridge
     Loaded: loaded (/lib/systemd/system/chirpstack-gateway-bridge.service; ena>
     Active: active (running) since Thu 2020-12-17 21:46:02 CET; 10min ago
       Docs: https://www.chirpstack.io/
   Main PID: 1470 (chirpstack-gate)
      Tasks: 9 (limit: 18927)
     Memory: 14.6M
     CGroup: /system.slice/chirpstack-gateway-bridge.service
             └─1470 /usr/bin/chirpstack-gateway-bridge
```

```
root@kol-GL65-Leopard-10SDR:/home/kol# sudo systemctl start chirpstack-applicat
on-server
root@kol-GL65-Leopard-10SDR:/home/kol# sudo systemctl status chirpstack-applica
ion-server
● chirpstack-application-server.service - ChirpStack Application Server
     Loaded: loaded (/lib/systemd/system/chirpstack-application-server.service;
     Active: active (running) since Thu 2020-12-17 21:46:02 CET; 12min ago
       Docs: https://www.chirpstack.io/
   Main PID: 1469 (chirpstack-appl)
      Tasks: 17 (limit: 18927)
     Memory: 41.1M
     CGroup: /system.slice/chirpstack-application-server.service
             └─1469 /usr/bin/chirpstack-application-server
```

```
root@kol-GL65-Leopard-10SDR:/home/kol# sudo systemctl start chirpstack-network-s
erver
root@kol-GL65-Leopard-10SDR:/home/kol# sudo systemctl status chirpstack-network-
server
● chirpstack-network-server.service - ChirpStack Network Server
     Loaded: loaded (/lib/systemd/system/chirpstack-network-server.service; ena>
     Active: active (running) since Thu 2020-12-17 21:46:02 CET; 14min ago
       Docs: https://www.chirpstack.io/
   Main PID: 1471 (chirpstack-netw)
      Tasks: 17 (limit: 18927)
     Memory: 28.3M
     CGroup: /system.slice/chirpstack-network-server.service
             └─1471 /usr/bin/chirpstack-network-server
```

Figure 39: Status of {Chirpstack Gateway bridge/ Chirpstack Application Server/ Chirpstack Network Server}

After that we go to the provided address “localhost:8080/” then connect to finally have our dashboard of ChirpStack working

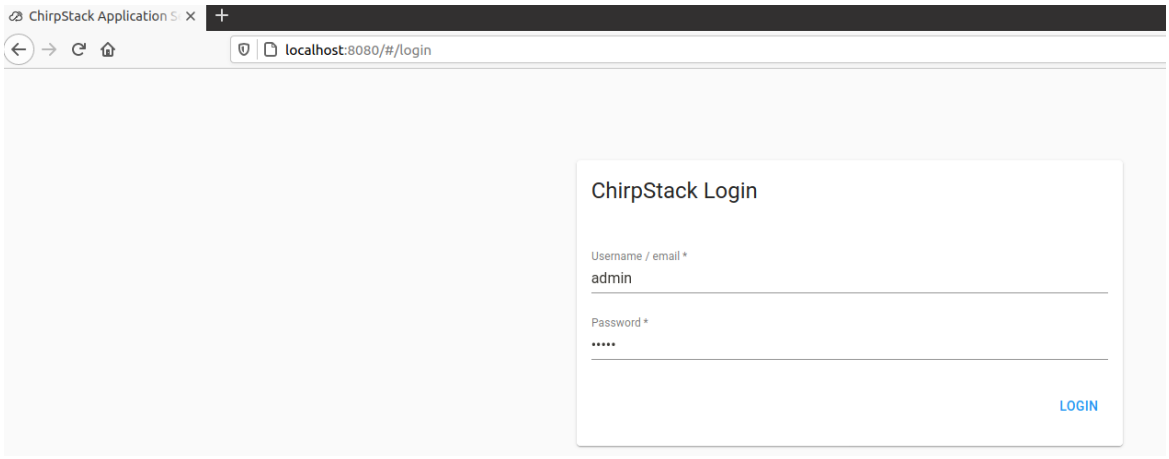


Figure 40: Chirpstack login

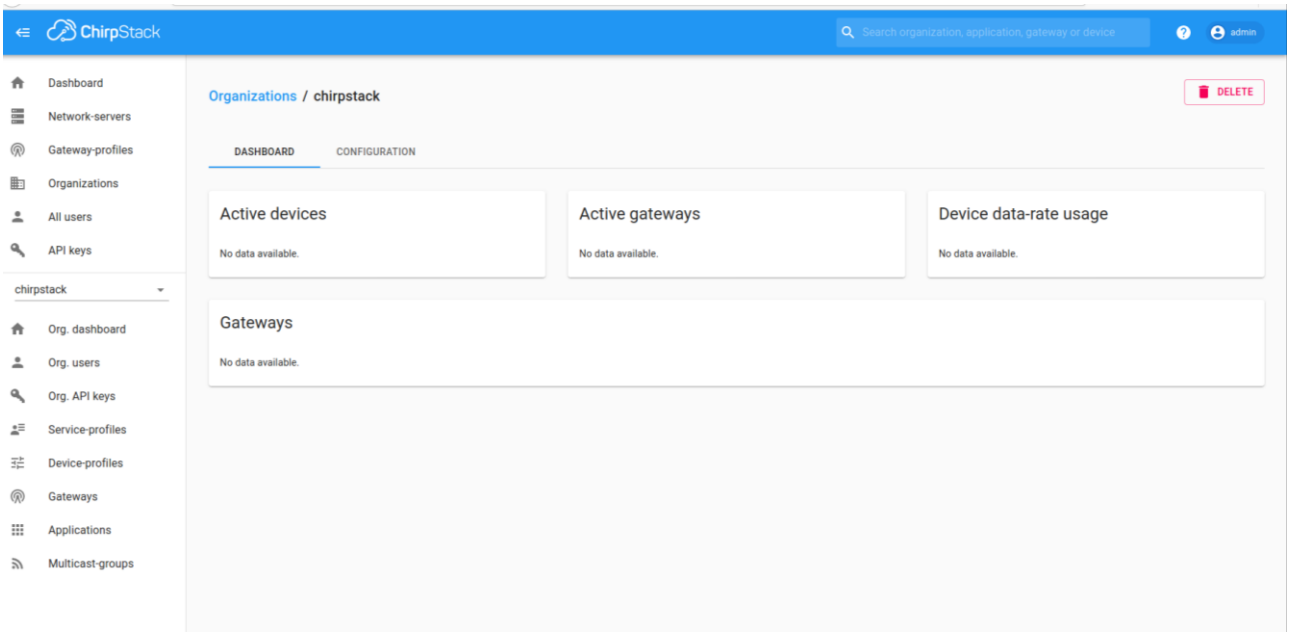


Figure 41: Chirpstack Dashboard

First we go to create a network sever it is mainly a default configuration

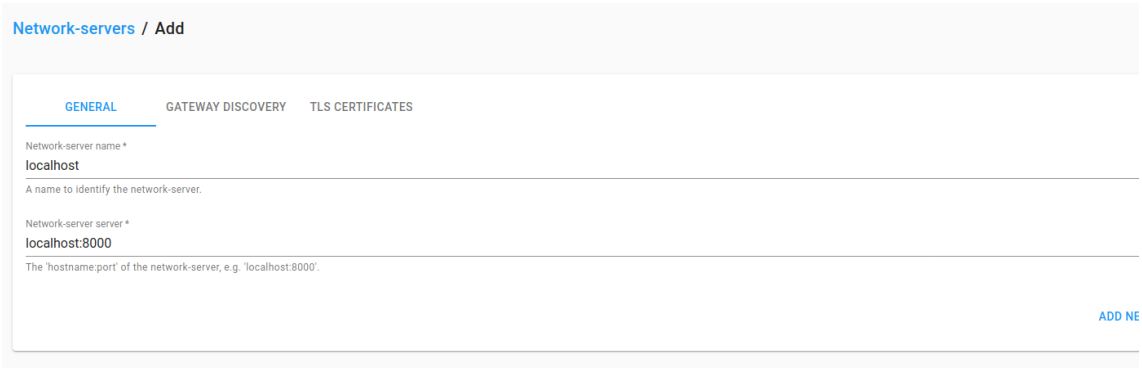


Figure 42: Used configuration (Network Server)

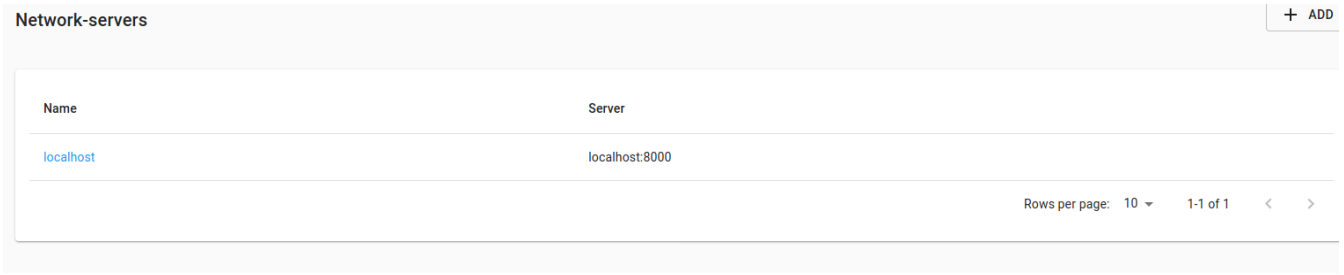


Figure 43: Created Network-server

Afterwards we go about creating a Gatway-profile which is basically the provided gateway configurations.

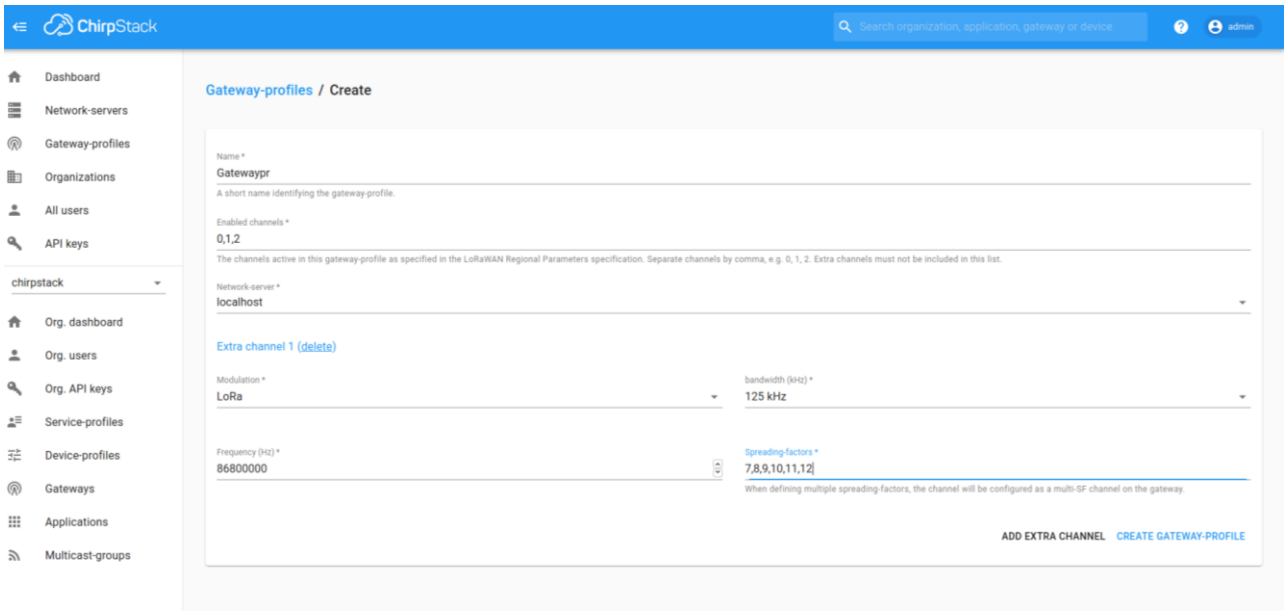


Figure 44: Used Configuration (Gateway Profile)

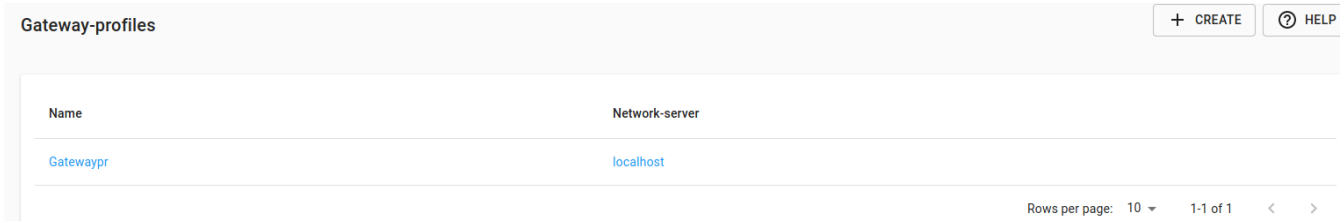


Figure 45: Created Gateway-profile

Now we go and configure our service-provider

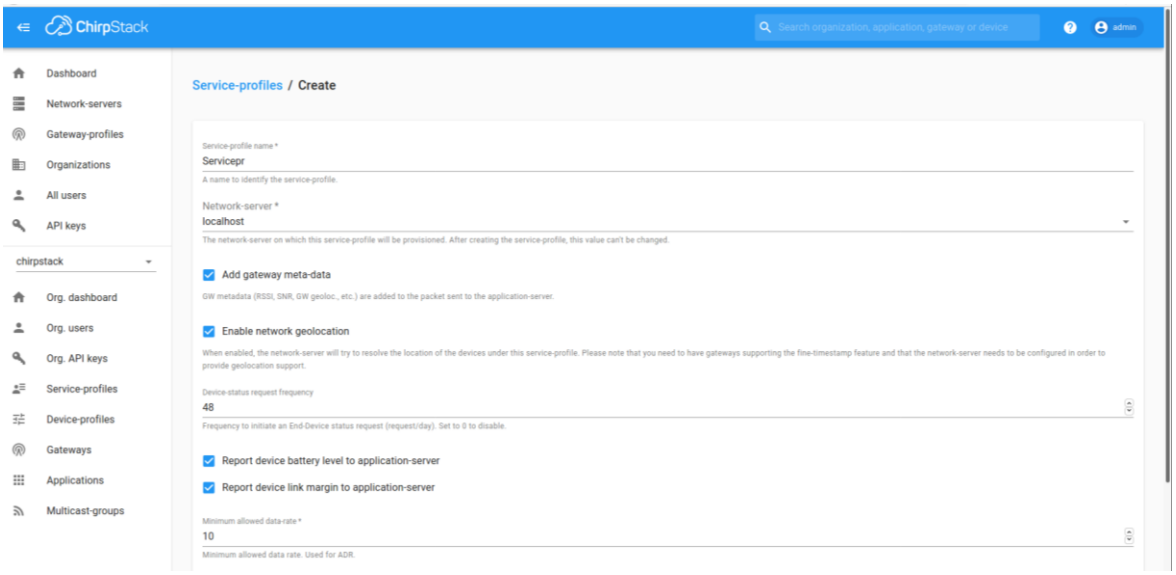


Figure 46: Used Configuration (Service-provider)

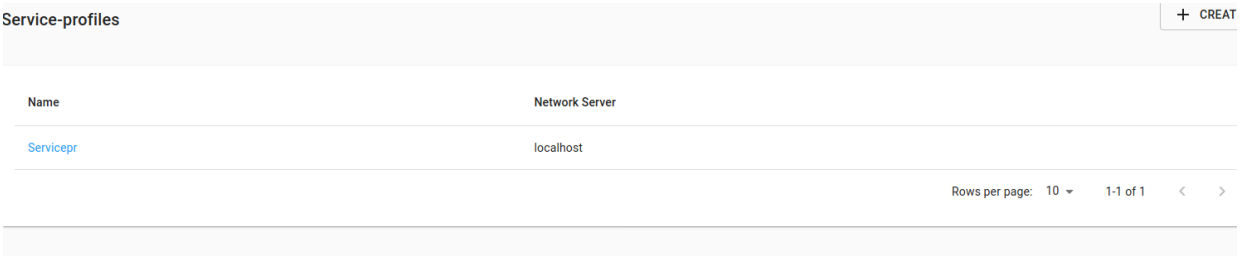


Figure 47: Created Service Provider

next I am creating the device-profile which is too basically the provided device configuration

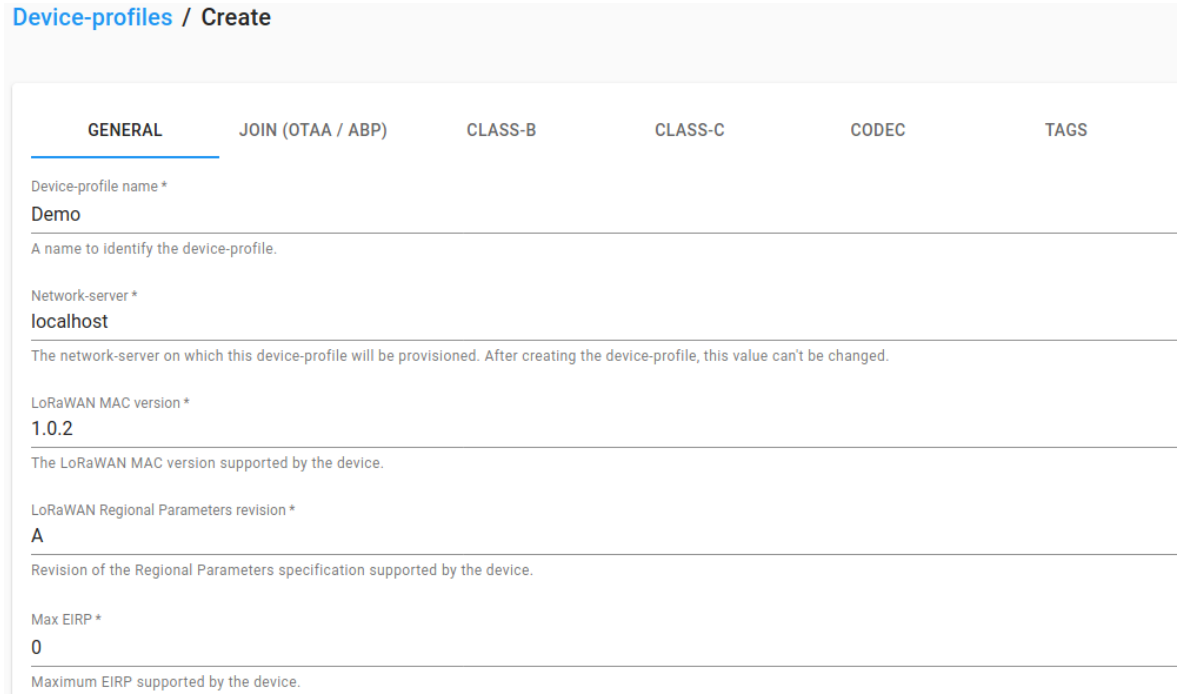


Figure 48: Used Configuration

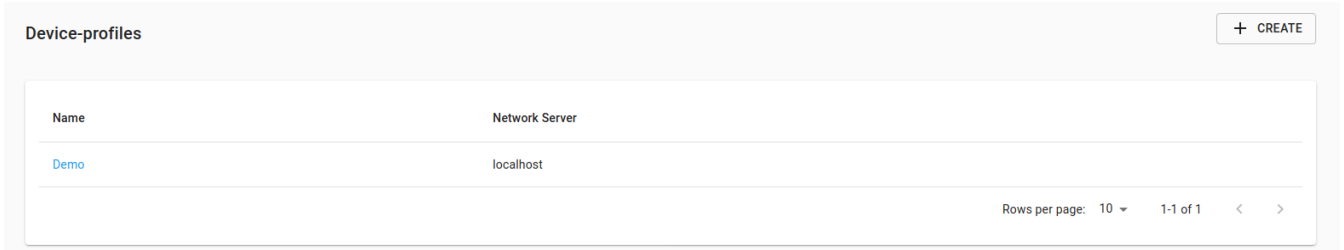


Figure 49: Created Device profile

Finally, we will be configuring our gateway, devices and applications

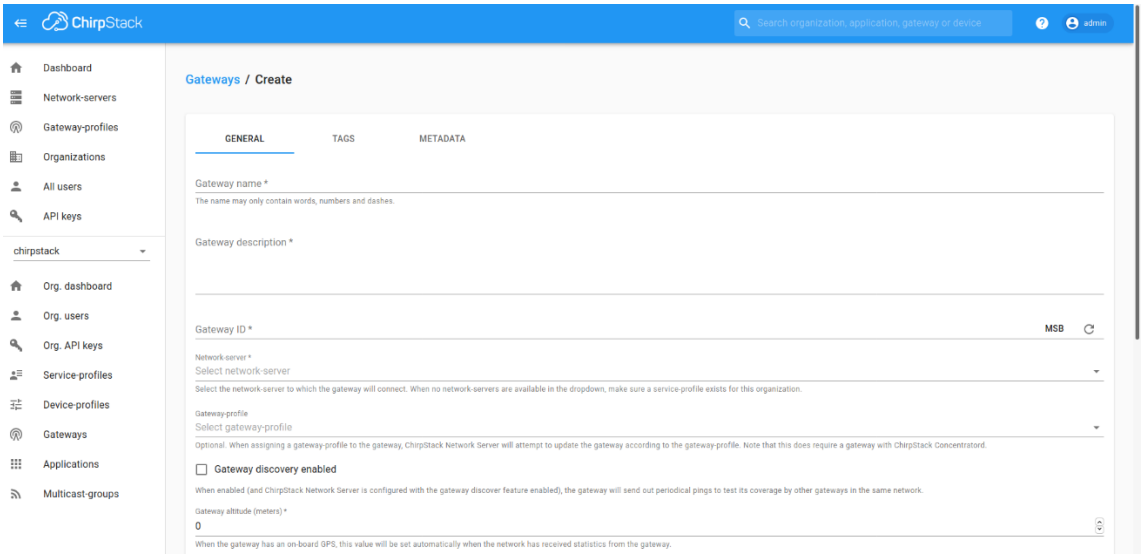


Figure 50: Gateway configuration(Gateway)

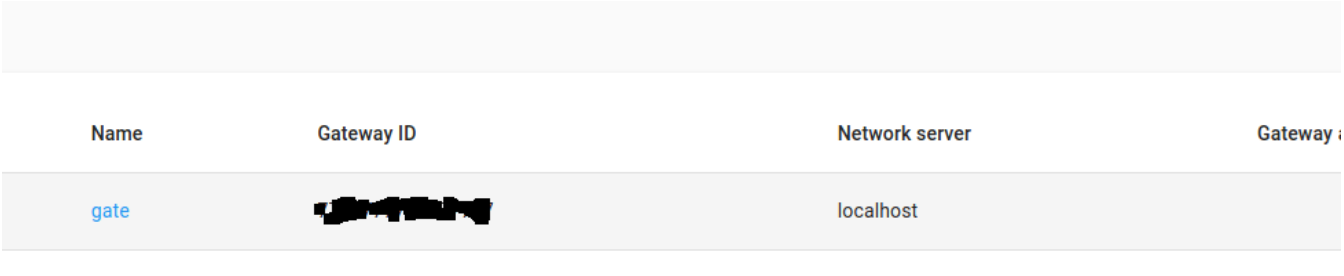


Figure 51: Created Gateway

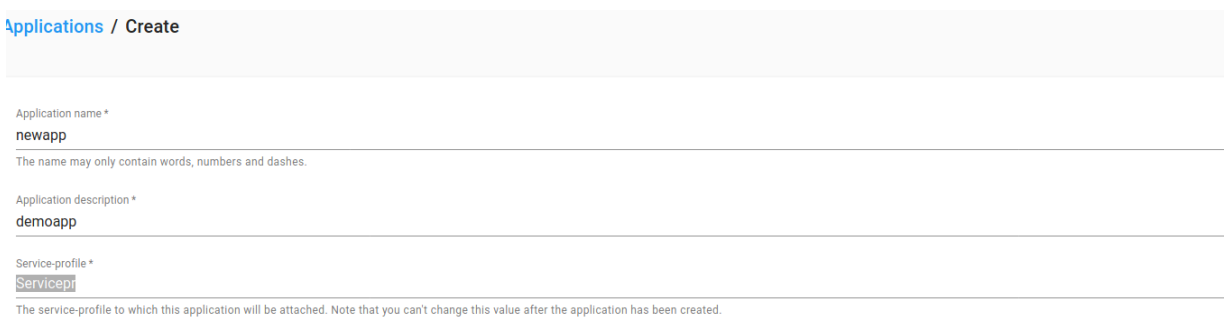


Figure 52: Used Configuration (application)

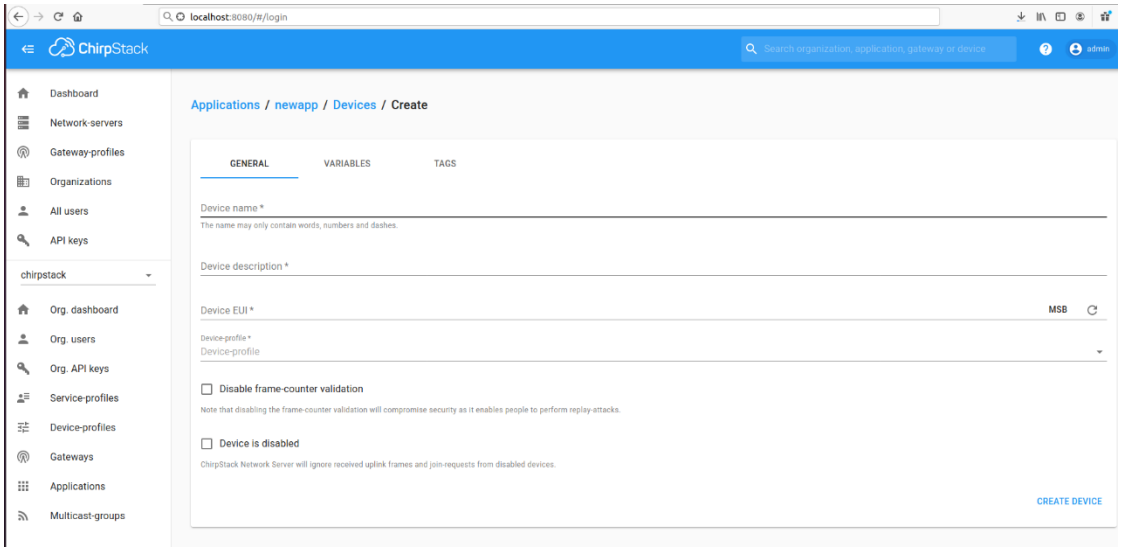


Figure 53: Device creation

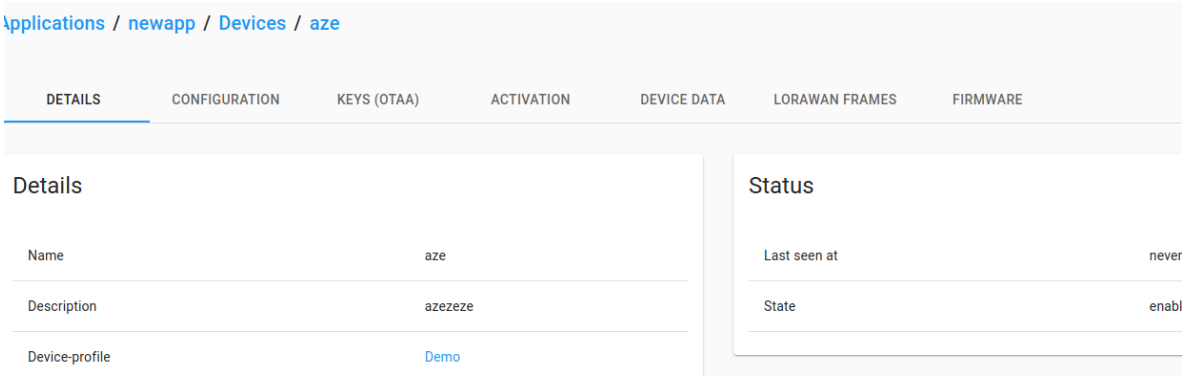


Figure 54: Device status

NOTE: The configured values were from the equipment configuration and default ones.

Finally, we can say that the connection is established after it finally detected the used equipment

Conclusion

In this chapter we detailed the project steps with the use of figures.

General Conclusion

In this report we have implemented the results of our work which is part of the telecommunication engineer internship at the Higher School of Communications of Tunis "SupCom". It is important to point out that this internship helped me while establishing the LoRaWAN connection using TTN and Chirpstack to encounter so many new kinds of technologies and thus have a better understanding of the way it works.
In this projects I have been tasked with the search and configure the LoRaWAN connection.
Thus, to achieve these objectives we began by putting the project in its general framework by presenting the host organization and detailing the objectives of this project. Then, we presented the various technologies and equipment essential to the realization of our project. Finally, we ended with the realization which allowed us to finally build the needed network.
In addition, this summer internship was very beneficial to me both technically and professionally.
I had the opportunity to enrich my knowledge on the IOT network and the cloud servers. Another rather important aspect of my experience was the opportunity to discover the professional environment with all that it implies of responsibility, discipline and seriousness.

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